If $|z| = 1$

$$|z^2 + z + 1| \leq |z|^2 + |z| + 1 = 3$$

$$|z^3 - 2| \geq \left| |z|^3 - |2| \right| = |1 - 2| = 1$$

If $|z| = 1/2$ $\qquad \left| \dfrac{1}{z^2 - 5z + 6} \right| \leq \dfrac{4}{15}$

$$\left| \frac{1}{z^2 - 5z + 6} \right| = \left| \frac{1}{(z-3)(z-2)} \right| = \frac{1}{|z-3|\,|z-2|}$$

$$\left| |z| - |w| \right| \leq |z - w|$$

$$|z - 3| \geq \left| |z| - |3| \right| = |1/2 - 3| = 5/2$$

$$|z - 2| \geq \left| |z| - |2| \right| = |1/2 - 2| = 2/3$$

$$\frac{1}{|z-3|} \leq \frac{2}{5} \qquad \frac{1}{|z-2|} \leq \frac{2}{3}$$

Remember:

From (8) with z_2 replaced by $-z_2$, we also find

$$|z_1 - z_2| \geq \left| |z_1| - |z_2| \right|. \tag{10}$$

In conclusion, we note that the triangle inequality (6) extends to any finite sum of complex numbers:

$$|z_1 + z_2 + z_3 + \cdots + z_n| \leq |z_1| + |z_2| + |z_3| + \cdots + |z_n|. \tag{11}$$

$$\frac{1}{|z-3| \cdot |z-2|} \leq \frac{2}{5} \cdot \frac{2}{3} = \frac{4}{15}$$

if $z = \dfrac{i}{3+i}$, $\bar{z} = ?$ $|z| = ?$

$$z = \frac{i}{3+i} = \frac{i\,(3-i)}{(3+i)(3-i)} = \frac{3i+1}{10} = \frac{1}{10} + i\frac{3}{10}$$

$$z = \frac{1}{10} + \frac{3i}{10} \Rightarrow \bar{z} = \frac{1}{10} - \frac{3i}{10}$$

$$|z| = \sqrt{\left(\frac{1}{10}\right)^2 + \left(-\frac{3}{10}\right)^2} = \sqrt{\frac{10}{100}}$$

$$= \frac{\sqrt{10}}{10} = \frac{1}{\sqrt{10}}$$

Show that the following equations are true.

a) $(1-i)^4 = -4$

b) $\dfrac{5}{(1-i)(2-i)(3-i)} = \dfrac{i}{2}$

c) $\overline{(\bar{z}+3i)} = z-3$

d) $\mathrm{Re}(iz) = -\mathrm{Im}\, z$

if $|z|=2$ $\qquad \left|\dfrac{-1}{z^4+3z^2+2}\right|$ Find an upper bound

$$\frac{|-1|}{|z^4+3z^2+2|} = \frac{1}{|z^4+3z^2+2|} \leq M \qquad M=?$$

$$|z^4+3z^2+2| = |z^2+1|\,|z^2+2| \geq ||z^2|-1|\cdot||z^2|-2|$$

$$||z_1|-|z_2|| \leq |z_1+z_2| \qquad |z|=2$$

$$|z^2+3z^2+2| \geq 3\cdot 2 = 6$$

$$\frac{1}{|z^2+3z^2+2|} \leq \frac{1}{6}$$

show the following inequality

Ex: $\sqrt{2}\,|z| \geq |\text{Re}z| + |\text{Im}z|$

$x = \text{Re}z$, $y = \text{Im}z$

$(|x| - |y|)^2 \geq 0$

$|x|^2 - 2|x||y| + |y|^2 \geq 0$

$|x|^2 + |y|^2 \geq 2|x||y|$ (add $|x|^2 + |y|^2$ to the both side)

$2(|x|^2 + |y|^2) \geq x^2 + 2|x||y| + |y|^2$

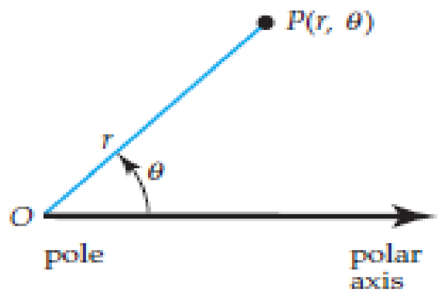
$2|z|^2 \geq (|x| + |y|)^2$

$\sqrt{2}\,|z| \geq |x| + |y| = |\text{Re}z| + |\text{Im}z|$

Ex: if $|z| = 1$; $|z^2 + z + 1| \leq |z|^2 + |z| + 1 = 3$

$|z^3 - 2| \geq ||z|^3 - |2|| = |1 - 2| = 1$

Polar Form of Complex Numbers



Polar coordinates

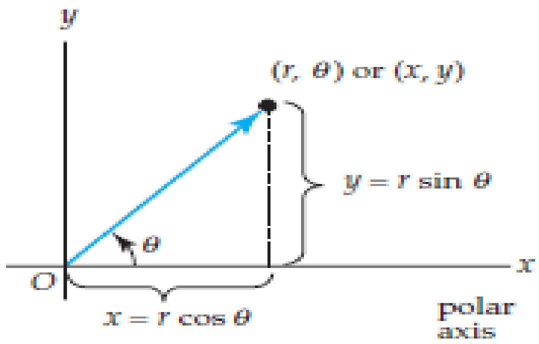
$$x = r\cos\theta,\ y = r\sin\theta.$$

$$z = x + iy \qquad z = (r\cos\theta) + i(r\sin\theta)$$

$$z = r(\cos\theta + i\sin\theta). \tag{1}$$

We say that (1) is the polar form or polar representation of the complex number z. Again, from Figure we see that the coordinate r can be interpreted as the distance from the origin to the point (x, y).

$$r = |z|.$$



Polar coordinates in the complex plane

The angle θ of inclination of the vector z, which *will always be measured in radians* from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos\theta = x/r$ and $\sin\theta = y/r$. An argument of a complex number z is not unique since $\cos\theta$ and $\sin\theta$ are 2π-periodic; in other words, if θ_0 is an argument of z, then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi, \dots$ are also arguments of z. In practice we use $\tan\theta = y/x$ to find θ. However, because $\tan\theta$ is π-periodic, some care must be exercised in using the last equation. A calculator will give only angles satisfying $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, that is, angles in the first and fourth quadrants. We have to choose θ consistent with the quadrant in which z is located; this may require adding or subtracting π to $\tan^{-1}(y/x)$ when appropriate. The following example illustrates how this is done.

Ex: Let $\arg w$ is defined $-\pi \leq w \leq \pi$ and

z is on $\underline{\underline{|z|=1}}$.

show that

$$\arg \frac{(z-1)}{(z+1)} = \begin{cases} \pi/2 & , \text{ if } \operatorname{Im} z > 0 \\ -\pi/2 & , \text{ if } \operatorname{Im} z < 0 \end{cases}$$

Let

$z = x+iy$; $x, y \in R$; on unit circle.

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy} = \frac{[(x-1)+iy][(x+1)-iy]}{[(x+1)+iy][(x+1)-iy]}$$

$$= \frac{(x-1)(x+1) - (x-1).yi + (x+1).iy + y^2}{(x+1)^2+y^2}$$

$$= \frac{x^2 - 1 + ((x+1)y - (x-1)y)i + y^2}{(x+1)^2+y^2}$$

$\downarrow$

$x^2+y^2=1$

$x^2+y^2=1$

$$\frac{z-1}{z+1} = \frac{2yi}{(x+1)^2+y^2} \Rightarrow \arg\left(\frac{z-1}{z+1}\right) = \begin{cases} \frac{\pi}{2} ; & y>0 \\ -\frac{\pi}{2} ; & y<0 \end{cases}$$

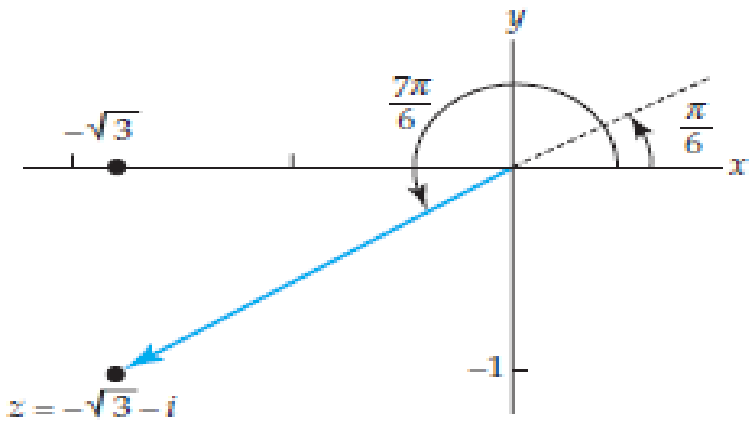
EXAMPLE 1 A Complex Number in Polar Form

Express $-\sqrt{3}-i$ in polar form.

Solution With $x=-\sqrt{3}$ and $y=-1$ we obtain $r=|z|=\sqrt{(-\sqrt{3})^2+(-1)^2}$ $=2$. Now $y/x=-1/(-\sqrt{3})=1/\sqrt{3}$, and so a calculator gives $\tan^{-1}(1/\sqrt{3})$ $=\pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $(-\sqrt{3},\ -1)$ lies in the third quadrant, we take the solution of $\tan\theta=-1/(-\sqrt{3})=1/\sqrt{3}$ to be $\theta=\arg(z)=\pi/6+\pi=7\pi/6$. See Figure

It follows from (1) that a polar form of the number is

$$z=2\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right). \tag{2}$$



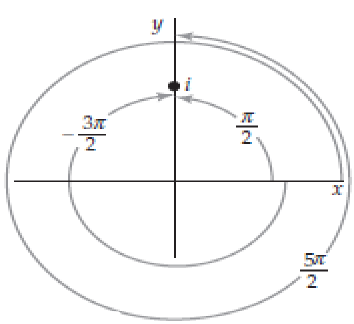
$\arg\left(-\sqrt{3}-i\right)$

Principal Argument The symbol $\arg(z)$ actually represents a set of values, but the argument θ of a complex number that lies in the interval $-\pi < \theta \leq \pi$ is called the **principal value** of $\arg(z)$ or the **principal argument** of z. The principal argument of z is *unique* and is represented by the symbol $\operatorname{Arg}(z)$, that is,

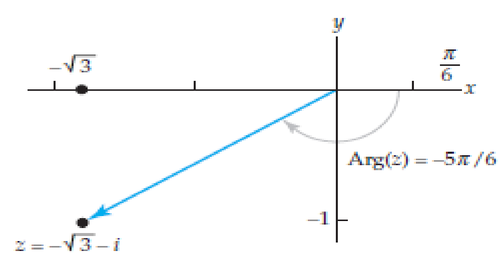
$$-\pi < \operatorname{Arg}(z) \leq \pi.$$

For example, if $z = i$, some values of $\arg(i)$ are $\pi/2$, $5\pi/2$, $-3\pi/2$, and so on, but $\operatorname{Arg}(i) = \pi/2$. Similarly, we see from Figure that the argument of $-\sqrt{3} - i$ that lies in the interval $(-\pi, \pi)$, the principal argument of z, is $\operatorname{Arg}(z) = \pi/6 - \pi = -5\pi/6$. Using $\operatorname{Arg}(z)$ we can express the complex number in (2) in the alternative polar form:

$$z = 2\left[\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right].$$



Some arguments of i



Principal argument

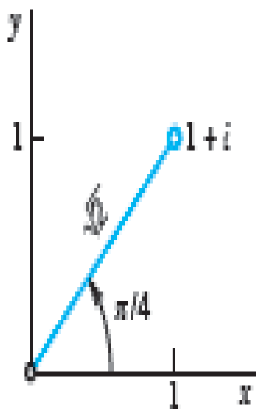
$z = -\sqrt{3} - i$

In general, $\arg(z)$ and $\operatorname{Arg}(z)$ are related by

$$\arg(z) = \operatorname{Arg}(z) + 2n\pi, \quad n = 0,\ \pm 1,\ \pm 2, \ldots. \tag{3}$$

For example, $\arg(i) = \dfrac{\pi}{2} + 2n\pi$. For the choices $n = 0$ and $n = -1$, (3) gives $\arg(i) = \operatorname{Arg}(i) = \pi/2$ and $\arg(i) = -3\pi/2$, respectively.

Polar Form of Complex Numbers. Principal Value Arg z



$z = 1 + i$ s the polar form $z = \sqrt{2}\,(\cos\frac{1}{4}\pi + i\sin\frac{1}{4}\pi)$. Hence we obtain

$$|z| = \sqrt{2}, \quad \arg z = \tfrac{1}{4}\pi \pm 2n\pi \ (n = 0, 1, \cdots), \qquad \text{and} \qquad \operatorname{Arg} z = \tfrac{1}{4}\pi \quad \text{(the principal value)}.$$

Similarly, $z = 3 + 3\sqrt{3}i = 6\,(\cos\frac{1}{3}\pi + i\sin\frac{1}{3}\pi)$, $|z| = 6$, and $\operatorname{Arg} z = \frac{1}{3}\pi$.

CAUTION! , we must pay attention to the quadrant in which z lies, since $\tan\theta$ has period π, so that the arguments of z and $-z$ have the same tangent. *Example:* for $\theta_1 = \arg(1 + i)$ and $\theta_2 = \arg(-1 - i)$ we have $\tan\theta_1 = \tan\theta_2 = 1$.

Multiplication and Division The polar form of a complex number is especially convenient when multiplying or dividing two complex numbers. Suppose

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \quad \text{and} \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

where θ_1 and θ_2 are any arguments of z_1 and z_2, respectively. Then

$$z_1 z_2 = r_1 r_2 \left[\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2\right)\right] \tag{4}$$

and, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 + i\left(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2\right)\right]. \tag{5}$$

From the addition formulas‡ for the cosine and sine, (4) and (5) can be rewritten as

$$z_1 z_2 = r_1 r_2\left[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\right] \tag{6}$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)\right]. \tag{7}$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2). \tag{8}$$

Proposition: Let $z_1 \neq 0$ and $z_2 \neq 0$

i) $\arg \overline{z_1} = -\arg z_1 \pmod{2\pi}$

$\overline{z_1} \cdot z_1$ is a real number. so,

$\arg(z_1 \cdot \overline{z_1}) = 0 \pmod{2\pi}$

$\arg(z_1) + \arg(\overline{z_1}) = 0 \pmod{2\pi}$

$\arg(\overline{z_1}) = -\arg(z_1) \pmod{2\pi}$

ii) $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 \pmod{2\pi}$

$\left(\frac{z_1}{z_2}\right) \cdot z_2 = z_1$ is true.

$\arg\left(\left(\frac{z_1}{z_2}\right) \cdot z_2\right) = \arg z_1 \pmod{2\pi}$

$\arg\left(\frac{z_1}{z_2}\right) + \arg(z_2) = \arg z_1 \pmod{2\pi}$

$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 \pmod{2\pi}$

iii) $\arg(1/z_1) = -\arg z_1 \pmod{2\pi}$

$$(1/z_1) \cdot z_1 = 1$$

$$\arg[(1/z_1) \cdot z_1] = \arg 1 \pmod{2\pi}$$

$$\arg(1/z_1) + \arg(z_1) = 0 \pmod{2\pi}$$

$$\arg(1/z_1) = -\arg(z_1)$$

EXAMPLE 2 Argument of a Product and of a Quotient

We have just seen that for $z_1 = i$ and $z_2 = -\sqrt{3} - i$ that $\mathrm{Arg}(z_1) = \pi/2$ an
$\mathrm{Arg}(z_2) = -5\pi/6$, respectively. Thus arguments for the product and quotien

$$z_1 z_2 = i(-\sqrt{3} - i) = 1 - \sqrt{3}i \quad \text{and} \quad \frac{z_1}{z_2} = \frac{i}{-\sqrt{3} - i} = -\frac{1}{4} - \frac{\sqrt{3}}{4}i$$

can be obtained from (8):

$$\arg(z_1 z_2) = \frac{\pi}{2} + \left(-\frac{5\pi}{6}\right) = -\frac{\pi}{3} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} - \left(-\frac{5\pi}{6}\right) = \frac{4\pi}{3}.$$

Integer Powers of z We can find integer powers of a complex number z from the results in (6) and (7). For example, if $z = r(\cos\theta + i\sin\theta)$, then with $z_1 = z_2 = z$, (6) gives

$$z^2 = r^2\left[\cos\left(\theta + \theta\right) + i\sin\left(\theta + \theta\right)\right] = r^2\left(\cos 2\theta + i\sin 2\theta\right).$$

Since $z^3 = z^2 z$, it then follows that

$$z^3 = r^3\left(\cos 3\theta + i\sin 3\theta\right),$$

and so on. In addition, if we take $\arg(1) = 0$, then (7) gives

$$\frac{1}{z^2} = z^{-2} = r^{-2}[\cos(-2\theta) + i\sin(-2\theta)].$$

Continuing in this manner, we obtain a formula for the nth power of z for any integer n:

$$z^n = r^n\left(\cos n\theta + i\sin n\theta\right). \tag{9}$$

When $n = 0$, we get the familiar result $z^0 = 1$.

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = -\sqrt{3} - i$.

Solution In (2) of Example 1 we saw that a polar form of the given number is $z = 2[\cos(7\pi/6) + i\sin(7\pi/6)]$. Using (9) with $r = 2$, $\theta = 7\pi/6$, and $n = 3$ we get

$$\left(-\sqrt{3} - i\right)^3 = 2^3\left[\cos\left(3\frac{7\pi}{6}\right) + i\sin\left(3\frac{7\pi}{6}\right)\right] = 8\left[\cos\frac{7\pi}{2} + i\sin\frac{7\pi}{2}\right] = -8i$$

since $\cos(7\pi/2) = 0$ and $\sin(7\pi/2) = -1$.

de Moivre's Formula When $z = \cos\theta + i\sin\theta$, we have $|z| = r = 1$, and so (9) yields

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta. \tag{10}$$

This last result is known as **de Moivre's formula** and is useful in deriving

EXAMPLE 4 de Moivre's Formula

From (10), with $\theta = \pi/6$, $\cos\theta = \sqrt{3}/2$ and $\sin\theta = 1/2$:

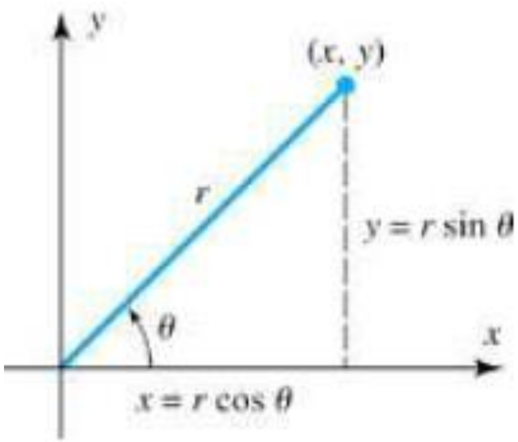
$$\begin{aligned}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3 &= \cos 3\theta + i \sin 3\theta = \cos\left(3 \cdot \frac{\pi}{6}\right) + i \sin\left(3 \cdot \frac{\pi}{6}\right)\\ &= \cos\frac{\pi}{2} + i \sin\frac{\pi}{2} = i.\end{aligned}$$

LET US REMEMBER

Polar Form

If $z = x + iy$ is a nonzero complex number, $r = |z|$, and θ measures the angle from the positive real axis to the vector z, then, as suggested by Figure

$$x = r\cos\theta, \qquad y = r\sin\theta$$



so that $z = x + iy$ can be written as $z = r\cos\theta + ir\sin\theta$ or

$$z = r(\cos\theta + i\sin\theta)$$

This is called a ***polar form of z***.

Argument of a Complex Number

The angle θ is called an ***argument of z*** and is denoted by

$$\theta = \arg z$$

The argument of z is not uniquely determined because we can add or subtract any multiple of 2π from θ to produce another value of the argument. However, there is only one value of the argument in radians that satisfies

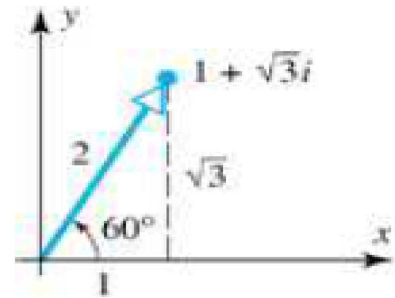
$$-\pi < \theta \leq \pi$$

This is called the ***principal argument of z*** and is denoted by

$$\theta = \operatorname{Arg} z$$

Express the following complex numbers in polar form using their principal arguments:

(a) $z = 1 + \sqrt{3}i$



(a)

The value of r is

$$r = |z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

and since $x = 1$ and $y = \sqrt{3}$, it follows from 1 that

$$1 = 2\cos\theta \qquad \text{and} \qquad \sqrt{3} = 2\sin\theta$$

so $\cos\theta = 1/2$ and $\sin\theta = \sqrt{3}/2$. The only value of θ that satisfies these relations and meets the requirement $-\pi < \theta \le \pi$ is $\theta = \pi/3 (= 60°)$ ( ). Thus a polar form of z is

$$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

(b) $z = -1 - i$

The value of r is

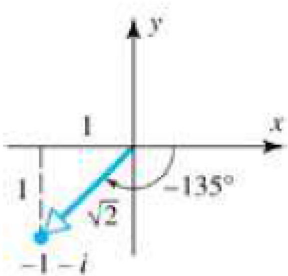
$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

and since $x = -1$, $y = -1$, it follows from 1 that

$$-1 = \sqrt{2}\cos\theta \quad \text{and} \quad -1 = \sqrt{2}\sin\theta$$

so $\cos\theta = -1/\sqrt{2}$ and $\sin\theta = -1/\sqrt{2}$. The only value of θ that satisfies these relations and meets the requirement $-\pi < \theta \leq \pi$ is $\theta = -3\pi/4 (= -135^{\circ})$. Thus, a polar form of z is

$$z = \sqrt{2}\left(\cos\frac{-3\pi}{4} + i\sin\frac{-3\pi}{4}\right)$$



(b)

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \quad \text{and} \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

$$z_1 z_2 = r_1 r_2 [(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + i(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)]$$

$$\cos(\theta_1 + \theta_2) = \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2$$
$$\sin(\theta_1 + \theta_2) = \sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2$$

$$\boxed{z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]}$$

if $z_2 \neq 0$, then

$$\boxed{\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]}$$

Let

$$z_1 = 1 + \sqrt{3}i \quad \text{and} \quad z_2 = \sqrt{3} + i$$

Polar forms of these complex numbers are

$$z_1 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \quad \text{and} \quad z_2 = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$\begin{aligned} z_1 z_2 &= 4\left[\cos\left(\frac{\pi}{3} + \frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right)\right] \\ &= 4\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right] = 4\,[0 + i] = 4i \end{aligned}$$

$$\begin{aligned} \frac{z_1}{z_2} &= 1 \cdot \left[\cos\left(\frac{\pi}{3} - \frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{3} - \frac{\pi}{6}\right)\right] \\ &= \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i \end{aligned}$$

As a check, we calculate $z_1 z_2$ and z_1 / z_2 directly without using polar forms for z_1 and z_2:

$$\begin{aligned} z_1 z_2 &= (1 + \sqrt{3}i)(\sqrt{3} + i) = (\sqrt{3} - \sqrt{3}) + (3 + 1)i = 4i \\ \frac{z_1}{z_2} &= \frac{1 + \sqrt{3}i}{\sqrt{3} + i} \cdot \frac{\sqrt{3} - i}{\sqrt{3} - i} = \frac{(\sqrt{3} + \sqrt{3}) + (-i + 3i)}{4} = \frac{\sqrt{3}}{2} + \frac{1}{2}i \end{aligned}$$

which agrees with our previous results.

DeMoivre's Formula

If n is a positive integer and $z = r(\cos\theta + i\sin\theta)$, then

$$z^n = \underbrace{z \cdot z \cdot z \cdots z}_{n-\text{factors}} = r^n[\cos\underbrace{(\theta + \theta + \cdots + \theta)}_{n-\text{terms}} + i\sin\underbrace{(\theta + \theta + \cdots + \theta)}_{n-\text{terms}}]$$

or

$$\boxed{z^n = r^n(\cos n\theta + i\sin n\theta)}$$

In the special case where $r = 1$, we have $z = \cos\theta + i\sin\theta$.

$$\boxed{(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta}$$

which is called ***DeMoivre's formula***.

Finding *n*th Roots

We now show how DeMoivre's formula can be used to obtain roots of complex numbers. If n is a positive integer and z is any complex number, then we define an ***nth root of z*** to be any complex number w that satisfies the equation

$$w^n = z \qquad *$$

We denote an nth root of z by $z^{1/n}$. If $z \neq 0$, then we can derive formulas for the nth roots of z as follows. Let

$$w = \rho(\cos\alpha + i\sin\alpha) \qquad \text{and} \qquad z = r(\cos\theta + i\sin\theta)$$

$$\rho^n(\cos n\alpha + i\sin n\alpha) = r(\cos\theta + i\sin\theta)$$

Comparing the moduli of the two sides, we see that $\rho^n = r$ or

$$\rho = \sqrt[n]{r}$$

where $\sqrt[n]{r}$ denotes the real positive nth root of r. Moreover, in order to have the equalities $\cos n\alpha = \cos\theta$ and $\sin n\alpha = \sin\theta$ the angles $n\alpha$ and θ must either be equal or differ by a multiple of 2π. That is,

$$n\alpha = \theta + 2k\pi \qquad \text{or} \qquad \alpha = \frac{\theta}{n} + \frac{2k\pi}{n}, \qquad k = 0, \pm 1, \pm 2, \ldots$$

Thus the values of $w = \rho(\cos\alpha + i\sin\alpha)$ that satisfy * are given by

$$w = \sqrt[n]{r}\left[\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i\sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)\right], \qquad k = 0, \pm 1, \pm 2, \ldots$$

Although there are infinitely many values of k, it can be shown ( ) that $k = 0, 1, 2, \ldots, n-1$ produce distinct values of w satisfying *but all other choices of k yield duplicates of these. Therefore, there are exactly n different nth roots of $z = r(\cos\theta + i\sin\theta)$, and these are given by

$$z^{1/n} = \sqrt[n]{r}\left[\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i\sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)\right], \qquad k = 0, 1, 2, \ldots, n-1$$

Find the three cube roots of $z = i$.

Solution Keep in mind that we are basically solving the equation $w^3 = i$. Now with $r = 1$, $\theta = \arg(i) = \pi/2$, a polar form of the given number is given by $z = \cos(\pi/2) + i\sin(\pi/2)$. with $n = 3$, we then obtain

$$w_k = \sqrt[3]{1}\left[\cos\left(\frac{\pi/2 + 2k\pi}{3}\right) + i\sin\left(\frac{\pi/2 + 2k\pi}{3}\right)\right], \quad k = 0, 1, 2.$$

Hence the three roots are,

$$k = 0, \quad w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

Find all cube roots of −8.

Solution

Since −8 lies on the negative real axis, we can use $\theta = \pi$ as an argument. Moreover, $r = |z| = |-8| = 8$, so a polar form of −8 is

$$-8 = 8(\cos\pi + i\sin\pi)$$

$n = 3$, it follows that

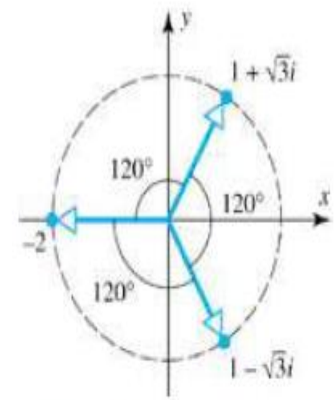
$$(-8)^{1/3} = \sqrt[3]{8}\left[\cos\left(\frac{\pi}{3} + \frac{2k\pi}{3}\right) + i\sin\left(\frac{\pi}{3} + \frac{2k\pi}{3}\right)\right], \qquad k = 0, 1, 2$$

Thus the cube roots of −8 are

$$2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 1 + \sqrt{3}i$$

$$2(\cos\pi + i\sin\pi) = 2(-1) = -2$$

$$2\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 1 - \sqrt{3}i$$



The cube roots of −8.

Fourth Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

Solution In this case, $r = \sqrt{2}$ and $\theta = \arg(z) = \pi/4$. $n = 4$, we obtain

$$w_k = \left(\sqrt{2}\right)^{1/4}\left[\cos\left(\frac{\pi/4 + 2k\pi}{4}\right) + i\sin\left(\frac{\pi/4 + 2k\pi}{4}\right)\right], \quad k = 0,\ 1,\ 2,\ 3.$$

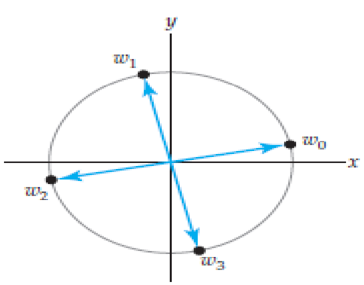
With the aid of a calculator we find

$$k = 0, \quad w_0 = \left(\sqrt{2}\right)^{1/4}\left[\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right] \approx 1.1664 + 0.2320i$$

$$k = 1, \quad w_1 = \left(\sqrt{2}\right)^{1/4}\left[\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right] \approx -0.2320 + 1.1664i$$

$$k = 2, \quad w_2 = \left(\sqrt{2}\right)^{1/4}\left[\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right] \approx -1.1664 - 0.2320i$$

$$k = 3, \quad w_3 = \left(\sqrt{2}\right)^{1/4}\left[\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right] \approx 0.2320 - 1.1664i.$$



3 Four fourth roots of $1 + i$

Geometrically, the n nth roots of a complex number z can also be interpreted as the vertices of a regular polygon with n sides that is inscribed within a circle of radius $\sqrt[n]{r}$ centered at the origin.

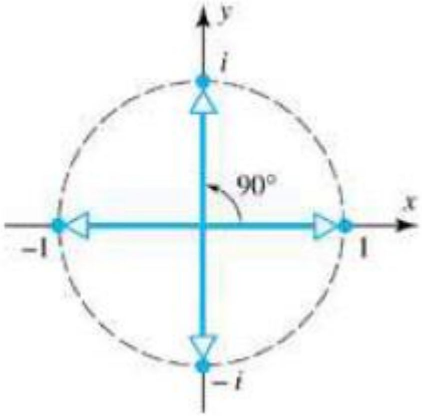
Find all fourth roots of 1.

Solution

$w = 1$ is one fourth root of 1, so the remaining three roots can be

generated by rotating this root through increments of $2\pi / 4 = \pi / 2$ radians $(= 90°)$.
roots of 1 are

$$1, \qquad i, \qquad -1, \qquad -i$$



$$z_1 = r_1 e^{i\theta_1} \qquad \text{and} \qquad z_2 = r_2 e^{i\theta_2}$$

are nonzero complex numbers, then

$$z_1 z_2 = r_1 r_2 e^{i\theta_1 + i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i\theta_1 - i\theta_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

We conclude this section with a useful formula for $\bar{z}$ in polar notation. If

$$\boxed{z = re^{i\theta} = r(\cos\theta + i\sin\theta)}$$

then

$$\bar{z} = r(\cos\theta - i\sin\theta)$$

Recalling the trigonometric identities

$$\sin(-\theta) = -\sin\theta \qquad \text{and} \qquad \cos(-\theta) = \cos\theta$$

$$\bar{z} = r[\cos(-\theta) + i\sin(-\theta)] = re^{i(-\theta)}$$

or, equivalently,

$$\bar{z} = re^{-i\theta}$$

In the special case where $r = 1$, the polar form of z is $z = e^{i\theta}$,

$$\overline{e^{i\theta}} = e^{-i\theta}$$

SOME EX.

$az^2 + bz + c = 0 \quad a, b, c \in R$

$a\left(z^2 + \frac{b}{a} z\right) + c = 0$

$a\left(\left(z + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right) + c = 0 \Rightarrow a\left(z + \frac{b}{2a}\right)^2 = \frac{b^2}{4a} - c$

$a\left(z + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a}$ ✓

$z + \frac{b}{2a} = \mp \frac{\sqrt{b^2 - 4ac}}{2a}$

$z = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$

$b^2 - 4ac < 0 \Rightarrow z = \frac{-b \mp \sqrt{\Delta}\sqrt{-1}}{2a} = \frac{-b \mp \sqrt{\Delta}\, i}{2a}$

(2) $\overset{a}{\underset{\sim}{i}} z^2 + \overbrace{(1+i)}^{b} z \overbrace{-2i}^{c} = 0$

$z^2 + \dfrac{(1+i)}{i} z - 2 = 0 \Rightarrow \overset{a}{1} \cdot z^2 + \overbrace{(1-i)}^{b} z \overbrace{-2}^{c} = 0$

$(-i)$

$$\underbrace{\left(z + \frac{(1-i)}{2}\right)^2 = 2 - \frac{i}{2}}$$

$\Downarrow$

$|2 - \frac{i}{2}| = \sqrt{4 + \frac{1}{4}} = \frac{\sqrt{17}}{2}$

$\mathrm{Arg}(2 - \frac{i}{2}) = \theta = \arctan\left(\frac{-1/2}{2}\right)$

$= \arctan(-1/4) = -14.036°$

$$z_k = -\frac{1+i}{2} + \left(\frac{\sqrt{17}}{2}\right)^{1/2} \left(\cos\frac{\theta + 2k\pi}{2} + i\sin\frac{\theta + 2k\pi}{2}\right)$$

$k = 0, 1$

(3) $z = (1+i)^n + (1-i)^n$

$|1+i| = \sqrt{2} \qquad Arg(1+i) = \frac{\pi}{4}$

$(1+i) = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

$(1+i)^n = (\sqrt{2})^n\left(\cos\frac{n\pi}{4} + i\sin n\frac{\pi}{4}\right)$

$(1-i)^n = (\sqrt{2})^n\left(\cos\frac{n\pi}{4} - i\sin n\frac{\pi}{4}\right)$

$z = 2(\sqrt{2})^n\left(\cos\frac{n\pi}{4}\right) = \bar{z}$

$|z| = 2(\sqrt{2})^n\left|\cos\frac{n\pi}{4}\right|$

show that

④ $z, w \in \mathbb{C} \Rightarrow |z+w|^2+|z-w|^2=2(|z|^2+|w|^2)$

$$\left.\begin{aligned} |z+w|^2 &= (z+w)(\overline{z+w}) = z\bar{z} + \cancel{z\bar{w}} + \cancel{w\bar{z}} + w\bar{w} \\ |z-w|^2 &= (z-w)(\overline{z-w}) = z\bar{z} - \cancel{z\bar{w}} - \cancel{w\bar{z}} + w\bar{w} \end{aligned}\right\} +$$

$$= 2(|z|^2+|w|^2)$$

⑤ for $\underbrace{a}_{\text{constant}} \in \mathbb{C}$; if $1-\bar{a}z \neq 0$ and $|z|=1$; $\left|\dfrac{z-a}{1-\bar{a}z}\right|=1$

$$\frac{|z-a|}{|1-\bar{a}z|} = \frac{|z-a|}{\underbrace{|\bar{z}|}_{|z|=1}|1-\bar{a}z|} = \frac{|z-a|}{|\bar{z}(1-\bar{a}z)|} = \frac{|z-a|}{(\bar{z}-\bar{a}|z|^2)} = \frac{|z-a|}{|\bar{z}-\bar{a}|}$$

$$= \frac{|z-a|}{|z-a|} = 1$$

$$z^{2/3} = (1-\sqrt{3}i)$$

$$2\pi - \frac{\pi}{3}$$

$$z^2 = (1-\sqrt{3}i)^3$$

$$1-\sqrt{3}i = 2 \cdot \left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)$$

$$r = \sqrt{1^2 + (\sqrt{3})^2}$$

$$r = 2$$

$$\operatorname{Arg}(1-\sqrt{3}i) = \theta = -\sqrt{3}$$

$$\Rightarrow z^2 = (1-\sqrt{3}i)^3 = (2)^3\left(\cos 3\cdot\frac{5\pi}{3} + i\sin 3\cdot\frac{5\pi}{3}\right)$$

$$z^2 = 8\cdot(\cos 5\pi + i\sin 5\pi)$$

$$z_k = 8^{1/2}\left(\cos\frac{5\pi+2k\pi}{2} + i\sin\frac{5\pi+2k\pi}{2}\right)$$

$$\underline{k = 0, 1}$$

$$2\pi + \frac{\pi}{2} = \frac{5\pi}{2}$$

$$3\pi + \frac{\pi}{2} = \frac{7\pi}{2}$$

$$z_0 = 2\sqrt{2}\,i$$

$$z_1 = -2\sqrt{2}\,i$$

$$|\alpha|<1 \implies |z|\le 1 \iff \frac{|z-\alpha|}{|1-\bar{\alpha}z|} \le 1$$

$$\left|\frac{z-\alpha}{1-\bar{\alpha}z}\right| \le 1 \iff |z-\alpha| \le |1-\bar{\alpha}z|$$

$$|z-\alpha|^2 \le |1-\bar{\alpha}z|^2$$

$$(z-\alpha)(\overline{z-\alpha}) \le (1-\bar{\alpha}z)(\overline{1-\bar{\alpha}z})$$

$$(z-\alpha)(\bar{z}-\bar{\alpha}) \le (1-\bar{\alpha}z)(1-\alpha\bar{z})$$

$$z.\bar{z} - \cancel{z\bar{\alpha}} - \cancel{\alpha\bar{z}} + \alpha\bar{\alpha} \le 1 - \cancel{\alpha\bar{z}} - \cancel{\bar{\alpha}z} + \bar{\alpha}\alpha z\bar{z}$$

$$|z|^2 + |\alpha|^2 \le 1 + |\alpha|^2|z|^2$$

$$|z|^2 - |\alpha|^2|z|^2 + |\alpha|^2 - 1 \le 0$$

$$|z|^2\left[1-|\alpha|^2\right] + \left[|\alpha|^2 - 1\right] \le 0$$

$$(|\alpha|^2-1)(1-|z|^2) \le 0$$

because $|\alpha|<1$

$|\alpha|^2-1<0$.

$$\Downarrow$$

$$\le 1$$

Circles Suppose $z_0 = x_0 + iy_0$. Since $|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ is the distance between the points $z = x + iy$ and $z_0 = x_0 + iy_0$, the points $z = x + iy$ that satisfy the equation

$$|z - z_0| = \rho, \quad \rho > 0, \tag{1}$$

lie on a **circle** of radius ρ centered at the point z_0.

(a) $|z| = 1$ is an equation of a unit circle centered at the origin.

(b) By rewriting $|z - 1 + 3i| = 5$ as $|z - (1 - 3i)| = 5$, we see from (1) that the equation describes a circle of radius 5 centered at the point $z_0 = 1 - 3i$.

$|z-4+3i|=5 \qquad z=a+ib\ ;\ a,b\in\mathbb{R}$

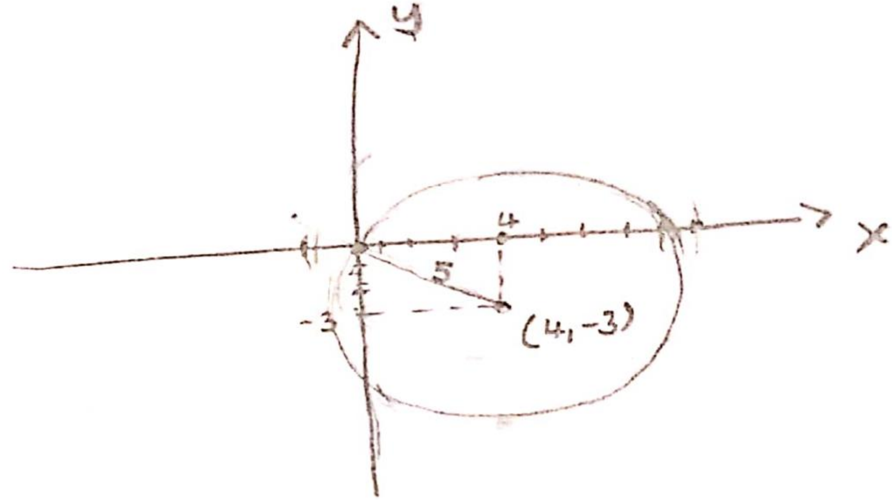
$|a+ib-4+3i|=5$

$|a-4+i(b+3)|=5$

$\sqrt{(a-4)^2+(b+3)^2}=5$

$(a-4)^2+(b+3)^2=25$

center: $(4,-3)$ $\quad r=5$



$\operatorname{Im}\left(\frac{1}{z}\right)=5 \implies \operatorname{Im}\left(\frac{1}{x+iy}\right)=\operatorname{Im}\left(\frac{x-iy}{(x+iy)(x-iy)}\right)=\frac{-y}{x^2+y^2}=5$

$z=x+iy,\ x,y\in\mathbb{R}$

$-y=5x^2+5y^2$

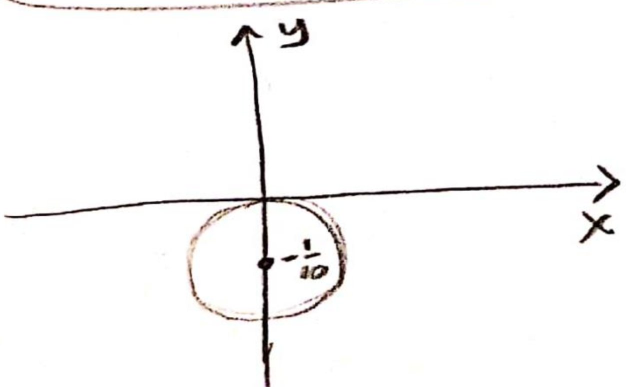
$5x^2+5y^2+y=0$

$5\left(x^2+y^2+\frac{y}{5}\right)=0$

$x^2+\left(y+\frac{1}{10}\right)^2-\frac{1}{100}=0$

$x^2+\left(y+\frac{1}{10}\right)^2=\frac{1}{100}$

center: $\left(0,-\frac{1}{10}\right)$ $\quad r=\frac{1}{10}$



if $\left|\frac{z-4}{z+4}\right| \le 3$; $z = a+ib$; $a,b \in \mathbb{R}$

$\left|\frac{a+ib-4}{a+ib+4}\right| \le 3 \Rightarrow |(a-4)+ib| \le 3\,|(a+4)+ib|$

$\sqrt{(a-4)^2+b^2} \le 3\sqrt{(a+4)^2+b^2}$

$(a-4)^2+b^2 \le 9[(a+4)^2+b^2]$

$a^2-8a+16+b^2 \le 9a^2+72a+9.16+9b^2$

$0 \le 8a^2+80a+8.16+8b^2$

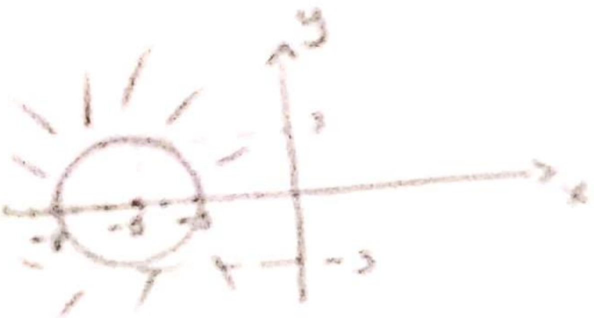
$0 \le a^2+10a+16+b^2$

$0 \le (a+5)^2+b^2-9$

$9 \le (a+5)^2+b^2$

center : $(-5,0)$

$r = 3$



if $|z-i|>1$

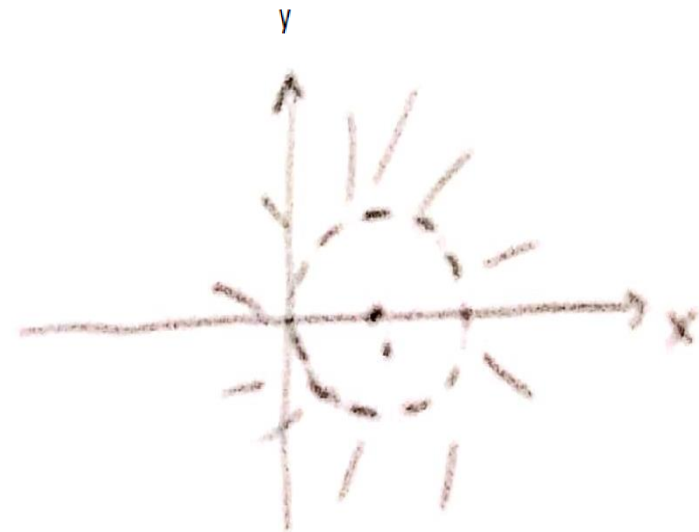
$|a+ib-i|>1$

$|a+i(b-1)|>1$

$\sqrt{a^2+(b-1)^2}>1$

$a^2+(b-1)^2>1$

center: $(0,1)$ $r=1$

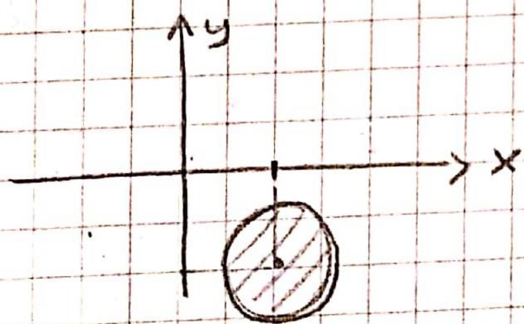


$|z-1+i| \leq 1/2$

$|z-(1-i)| \leq 1/2$

$|(x-1)-i(y+1)| \leq 1/2$

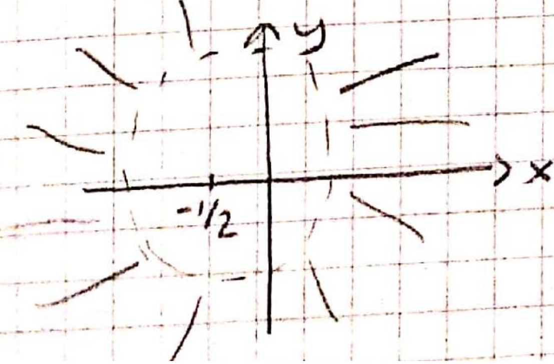
$(x-1)^2+(y+1)^2 \leq 1/4$ $(1,-1) \rightarrow$ center $r=1/2 \rightarrow$ radius



$|2z+1| > 2$

$|2z+1| = 2|z+1/2| > 2 \Rightarrow |z+1/2| > 1$ $(-1/2, 0)$

$r=1$



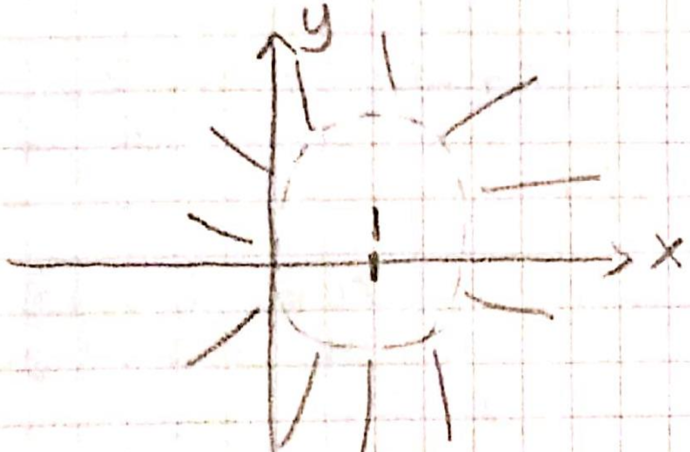
$2\,\mathrm{Re}\,z < |z|^2$

$2x < x^2+y^2 \Rightarrow 0 < x^2-2x+y^2+1-1$

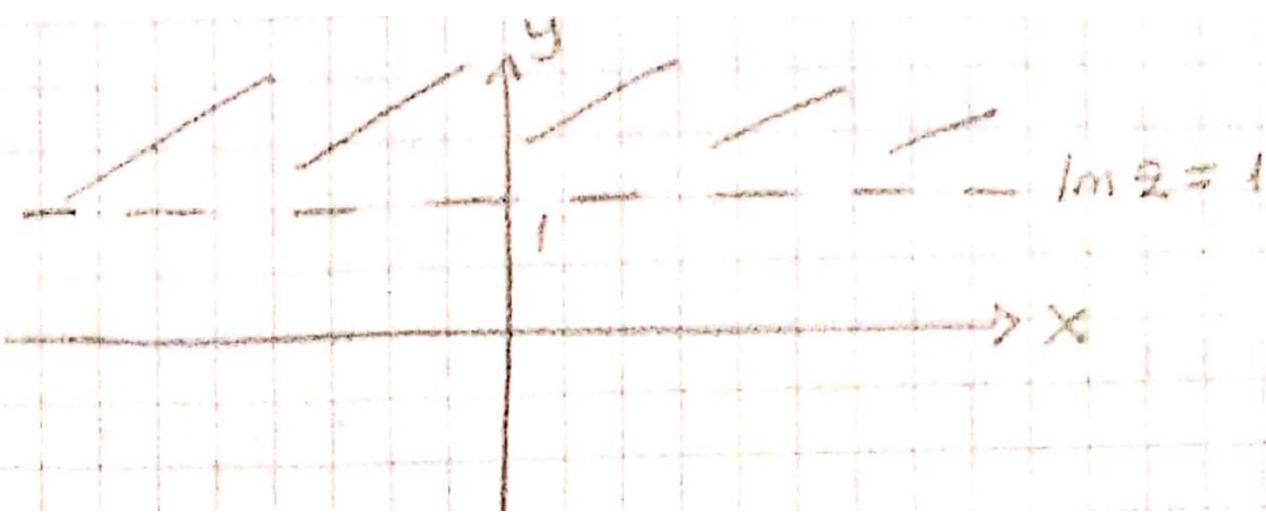
$1 < (x-1)^2+y^2$

$(1,0) \to$ center

$r = 1 \to$ radius



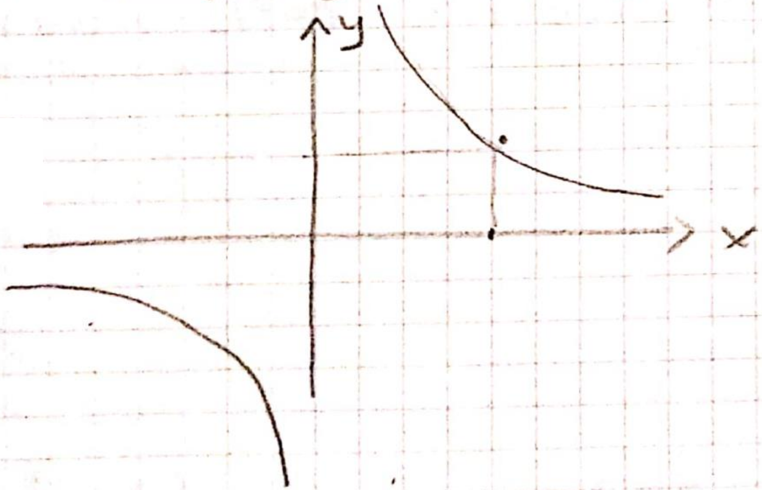
$\mathrm{Im}\,z > 1$



$\operatorname{Im}(z^2) = 4$

$z^2 = (x+iy)^2 \quad x^2 - y^2 + 2ixy \Rightarrow \quad \operatorname{Im} z^2 = 2xy = 4$

$xy = 2$



$|\sqrt{3} - i| = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$

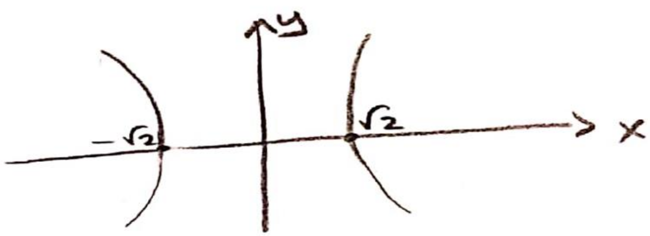
$\operatorname{Re}(z^2) = |\sqrt{3} - i|$

$z = a + ib \ ; \quad a, b \in \mathbb{R}$

$\operatorname{Re}((a+ib)^2) = \operatorname{Re}(a^2 + 2iab - b^2) = |\sqrt{3} - i|$

$a^2 - b^2 = 2 \quad \Longrightarrow \quad \dfrac{a^2}{(\sqrt{2})^2} - \dfrac{b^2}{(\sqrt{2})^2} = 1$

hyperbola

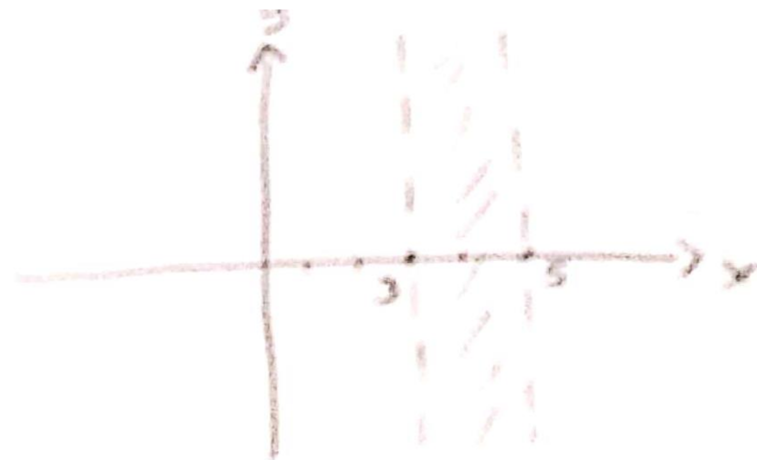


1F $2 < Re(z-1) < 4$

$2 < Re(x+iy-1) < 4$

$2 < x-1 < 4$

$3 < x < 5$



1F $|z-i| > 1$

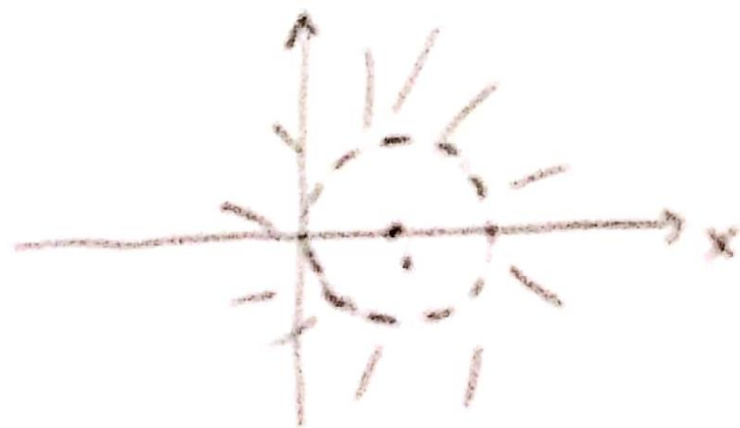
$|a+ib-i| > 1$

$|a+i(b-1)| > 1$

$\sqrt{a^2+(b-1)^2} > 1$

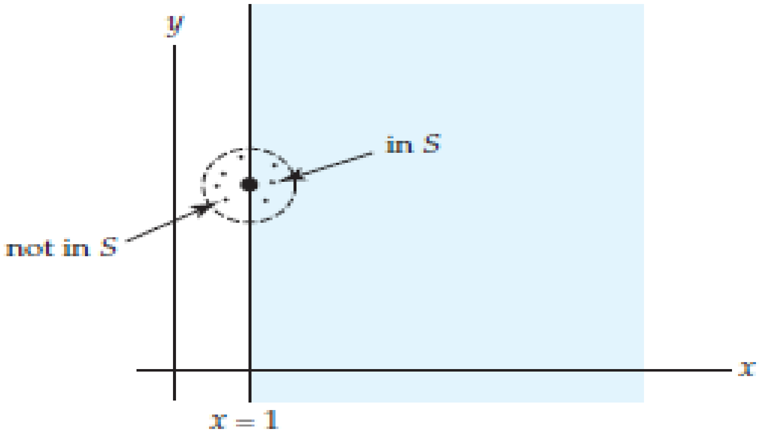
$a^2+(b-1)^2 > 1$

center: $(0,1)$ $r=1$



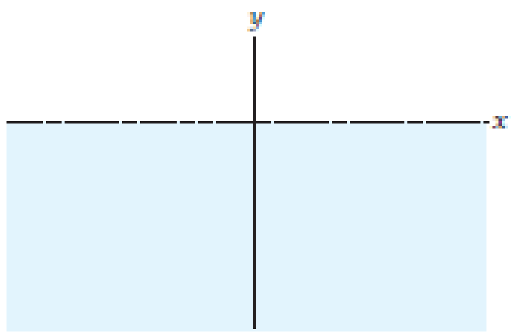
Disks and Neighborhoods The points z that satisfy the inequality $|z - z_0| \leq \rho$ can be either *on* the circle $|z - z_0| = \rho$ or *within* the circle. We say that the set of points defined by $|z - z_0| \leq \rho$ is a **disk** of radius ρ centered at z_0. But the points z that satisfy the strict inequality $|z - z_0| < \rho$ lie within, and not on, a circle of radius ρ centered at the point z_0. This set is called a **neighborhood** of z_0. Occasionally, we will need to use a neighborhood of z_0 that also *excludes* z_0. Such a neighborhood is defined by the simultaneous inequality $0 < |z - z_0| < \rho$ and is called a **deleted neighborhood** of z_0. For example, $|z| < 1$ defines a neighborhood of the origin, whereas $0 < |z| < 1$ defines a deleted neighborhood of the origin; $|z - 3 + 4i| < 0.01$ defines a neighborhood of $3 - 4i$, whereas the inequality $0 < |z - 3 + 4i| < 0.01$ defines a deleted neighborhood of $3 - 4i$.

Open Sets A point z_0 is said to be an **interior point** of a set S of the complex plane if there exists some neighborhood of z_0 that lies entirely within S. If every point z of a set S is an interior point, then S is said to be an **open set**. For example, the inequality $\mathrm{Re}(z) > 1$ defines a *right half-plane*, which is an open set.

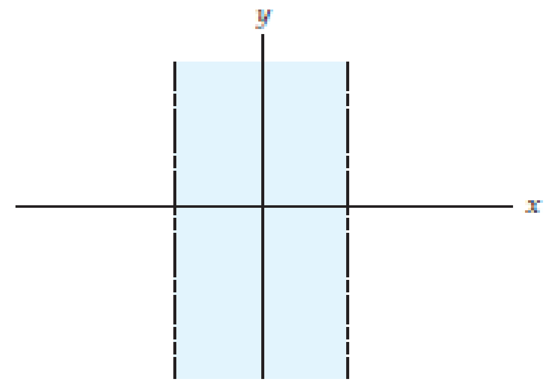


Set S not open

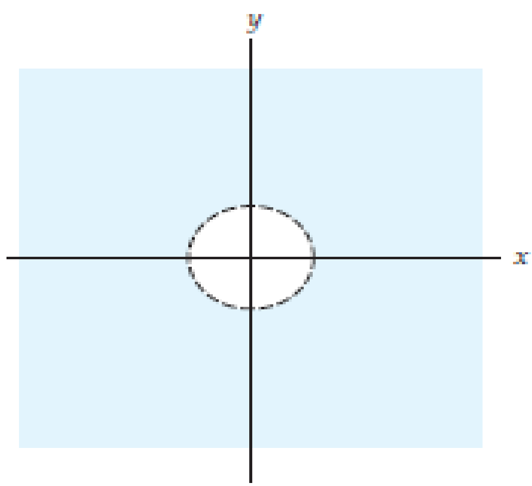
Some Open Sets



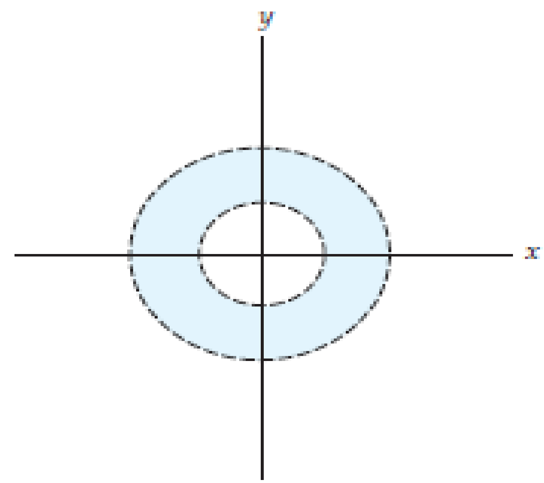
(a) Im(z) < 0; lower half-plane



(b) –1 < Re(z) < 1; infinite vertical strip

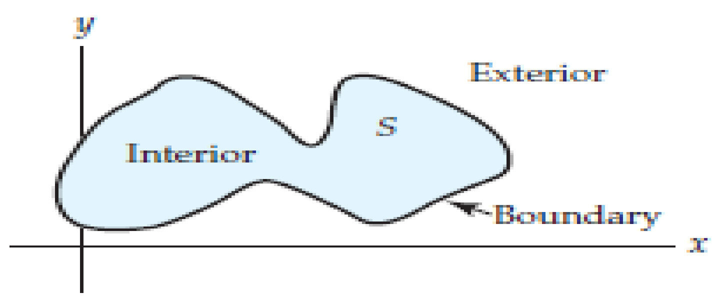


(c) $|z| > 1$; exterior of unit circle



(d) $1 < |z| < 2$; interior of circular ring

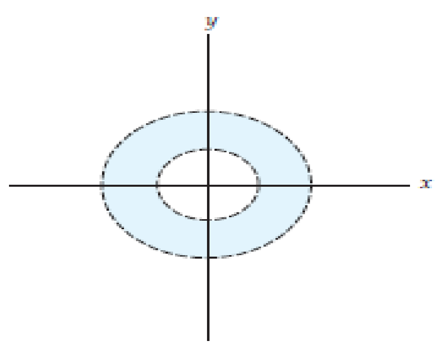
If *every* neighborhood of a point z_0 of a set S contains at least one point of S and at least one point not in S, then z_0 is said to be a **boundary point** of S. For the set of points defined by $\text{Re}(z) \geq 1$, the points on the vertical line $x = 1$ are boundary points. The points that lie on the circle $|z - i| = 2$ are boundary points for the disk $|z - i| \leq 2$ as well as for the neighborhood $|z - i| < 2$ of $z = i$. The collection of boundary points of a set S is called the **boundary** of S. The circle $|z - i| = 2$ is the boundary for both the disk $|z - i| \leq 2$ and the neighborhood $|z - i| < 2$ of $z = i$. A point z that is neither an interior point nor a boundary point of a set S is said to be an **exterior point** of S; in other words, z_0 is an exterior point of a set S if there exists *some* neighborhood of z_0 that contains no points of S.



Annulus The set S_1 of points satisfying the inequality $\rho_1 < |z - z_0|$ lie exterior to the circle of radius ρ_1 centered at z_0, whereas the set S_2 of points satisfying $|z - z_0| < \rho_2$ lie interior to the circle of radius ρ_2 centered at z_0. Thus, if $0 < \rho_1 < \rho_2$, the set of points satisfying the simultaneous inequality

$$\rho_1 < |z - z_0| < \rho_2, \tag{2}$$

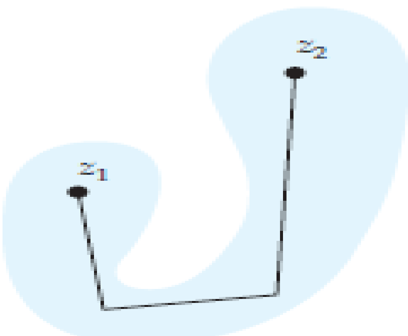
is the intersection of the sets S_1 and S_2. This intersection is an open circular ring centered at z_0. Figure illustrates such a ring centered at the origin. The set defined by (2) is called an open **circular annulus**. By allowing $\rho_1 = 0$, we obtain a deleted neighborhood of z_0.



$1 < |z| < 2$; interior of circular ring

Domain If any pair of points z_1 and z_2 in a set S can be connected by a polygonal line that consists of a finite number of line segments joined end to end that lies entirely in the set, then the set S is said to be **connected**.

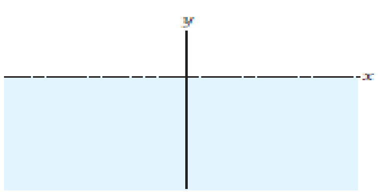
An open connected set is called a **domain**.



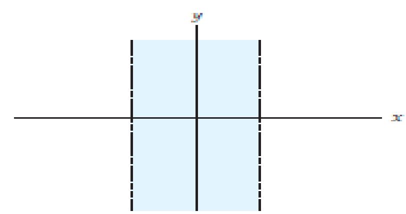
Connected set

Regions A region is a set of points in the complex plane with all, some, or none of its boundary points. Since an open set does not contain any boundary points, it is automatically a region. A region that contains all its boundary points is said to be **closed**. The disk defined by $|z - z_0| \leq \rho$ is an example of a closed region and is referred to as a **closed disk**. A neighborhood of a point z_0 defined by $|z - z_0| < \rho$ is an open set or an open region and is said to be an **open disk**. If the center z_0 is deleted from either a closed disk or an open disk, the regions defined by $0 < |z - z_0| \leq \rho$ or $0 < |z - z_0| < \rho$ are called **punctured disks**.

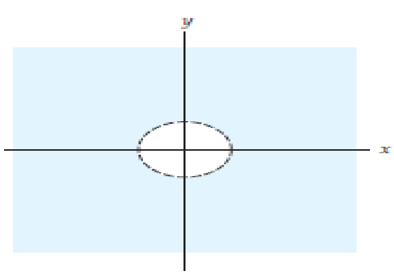
Bounded Sets Finally, we say that a set S in the complex plane is **bounded** if there exists a real number $R > 0$ such that $|z| < R$ every z in S. That is, S is bounded if it can be completely enclosed within *some* neighborhood of the origin. In Figure 1. the set S shown in color is bounded because it is contained entirely within the dashed circular neighborhood of the origin. A set is **unbounded** if it is not bounded. For example, the set in Figure (d) is bounded, whereas the sets in Figures (a), (b), and (c) are unbounded.



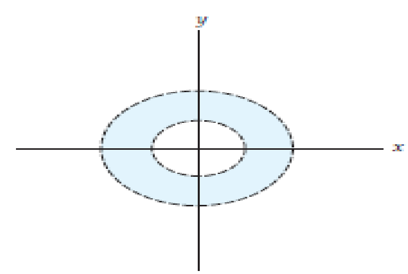
(a) Im(z) < 0; lower half-plane



(b) –1 < Re(z) < 1; infinite vertical strip



(c) |z| > 1; exterior of unit circle



(d) 1 < |z| < 2; interior of circular ring

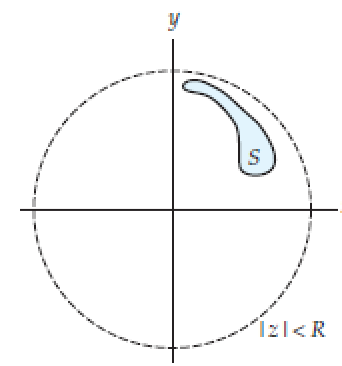
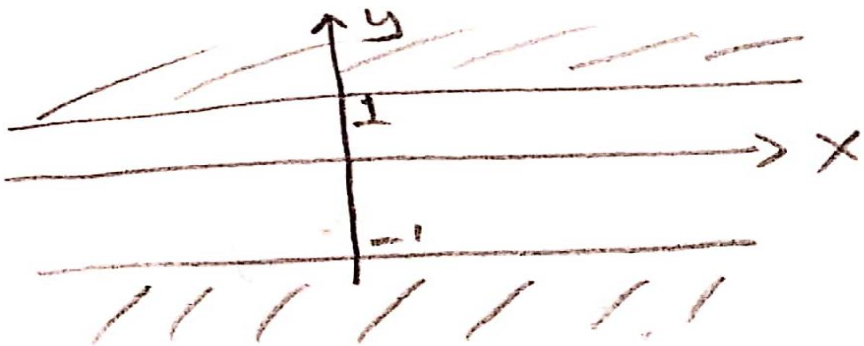


Figure 1. The set S is bounded since some neighborhood of the origin encloses S entirely.

$S: |\operatorname{Im} z| > 1$ is it connected?

$\operatorname{Im} z < -1 \qquad \operatorname{Im} z > 1$

$\operatorname{Im} z = y \qquad y > 1 \qquad y < -1$



It is not connected

whether $|z-4| \geq |z|$ shows a domain ?

$|z-4| \geq |z|$

$|x+iy-4| \geq |x+iy|$

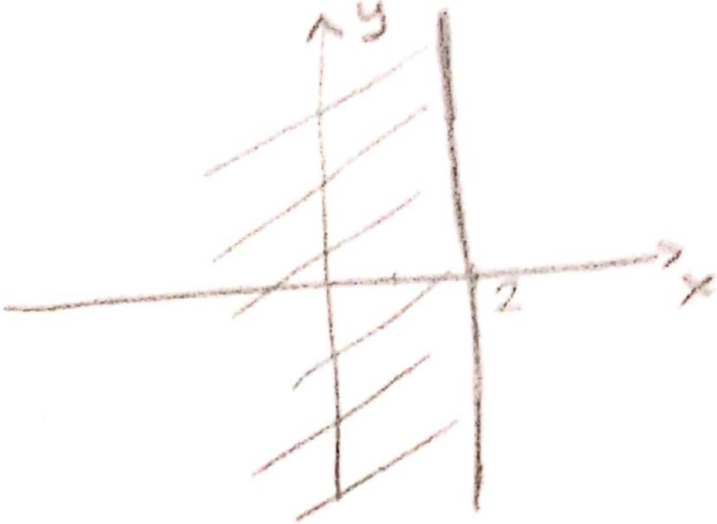
$\sqrt{(x-4)^2+y^2} \geq \sqrt{x^2+y^2}$

$(x-4)^2+y^2 \geq x^2+y^2$

$\cancel{x^2}-8x+16+\cancel{y^2} \geq \cancel{x^2}+\cancel{y^2}$

$16 \geq 8x$

$2 \geq x$



İt is not open

(It is connected

<u>İt is not domain</u>

additional information on definitions

By a **point set** in the complex plane we mean any sort of collection of finitely many or infinitely many points. Examples are the solutions of a quadratic equation, the points of a line, the points in the interior of a circle as well as the sets discussed just before.

A set S is called **open** if every point of S has a neighborhood consisting entirely of points that belong to S. For example, the points in the interior of a circle or a square form an open set, and so do the points of the right half-plane $\text{Re } z = x > 0$.

A set S is called **connected** if any two of its points can be joined by a chain of finitely many straight-line segments all of whose points belong to S. An open and connected set is called a **domain.** Thus an open disk and an open annulus are domains. An open square with a diagonal removed is not a domain since this set is not connected. (Why)

The **complement** of a set S in the complex plane is the set of all points of the complex plane that ***do not elong*** to S. A set S is called **closed** if its complement is open. For example, the points on and inside the unit circle form a closed set ("closed unit disk") since its complement $z| > 1$ is open.

A **boundary point** of a set S is a point every neighborhood of which contains both points that belong to S and points that do not belong to S. For example, the boundary points of an annulus are the points on the two bounding circles. Clearly, if a set S is open, then no boundary point belongs to S; if S is closed, then every boundary point belongs to S. The set of all boundary points of a set S is called the **boundary** of S.

A **region** is a set consisting of a domain plus, perhaps, some or all of its boundary points.

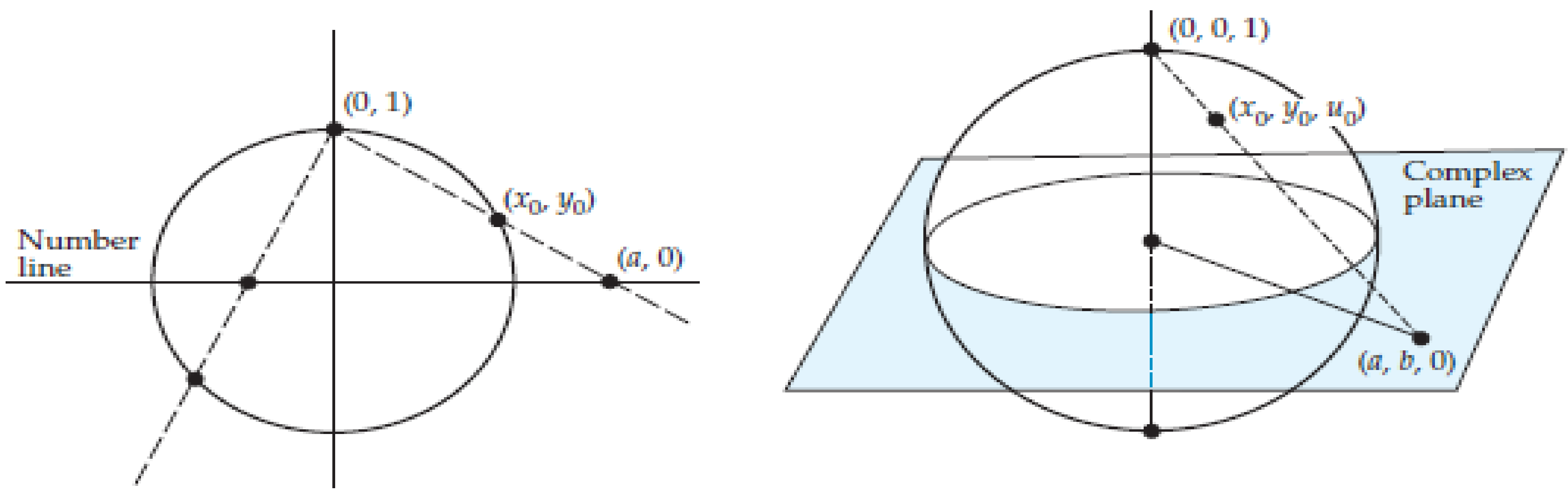
WARNING "Domain" is the *modern* term for an open connected set. Nevertheless, some authors still call a domain a "region" and others make no distinction between the two terms.

For example, in a course in calculus you dealt with *limits at infinity*, where the behavior of functions was examined as x either increased or decreased without bound. Since we have exactly two directions on a number line, it is convenient to represent the notions of "increasing without bound" and "decreasing without bound" symbolically by $x \to +\infty$ and $x \to -\infty$, respectively. It turns out that we can get along just fine without the designation $\pm\infty$ by dealing with an "ideal point" called the **point at infinity,** which is denoted simply by ∞. To do this we identify any real number a with a point (x_0, y_0) on a unit circle $x^2 + y^2 = 1$ by drawing a straight line from the point $(a, 0)$ on the x-axis or horizontal number line to the point $(0, 1)$ on the circle. The point (x_0, y_0) is the point of intersection of the line with the circle.

The only point on the circle that does not correspond to a real number a is $(0, 1)$. To complete the correspondence with *all* the points on the circle, we identify $(0, 1)$ with ∞. The set consisting of the real numbers $\mathbf{R}$ adjoined with ∞ is called the **extended real-number system.**

Recall that since $\mathbf{C}$ is not ordered, the notions of z either "increasing" or "decreasing" have no meaning. However, we know that by increasing the modulus $|z|$ of a complex number z, the number moves farther from the origin. If we allow $|z|$ to become unbounded, say, along the real and imaginary axes, we do not have to distinguish between "directions" on these axes by notations such as $z \to +\infty$, $z \to -\infty$, $z \to +i\infty$, and $z \to -i\infty$. In complex analysis, only the notion of ∞ is used because we can extend the complex number system $\mathbf{C}$ in a manner analogous to that just described for the real number system $\mathbf{R}$. This time, however, we associate a complex number with a point on a unit sphere called the **Riemann sphere.** By drawing a line from the number $z = a + ib$, written as $(a, b, 0)$, in the complex plane to the north pole $(0, 0, 1)$ of the sphere $x^2 + y^2 + u^2 = 1$, we determine a unique point (x_0, y_0, u_0) on a unit sphere. a complex number with a very large modulus is far from the origin $(0, 0, 0)$ and, correspondingly, the point (x_0, y_0, u_0) is close to $(0, 0, 1)$. In this manner each complex number is identified with a single point on the sphere. Because the point $(0, 0, 1)$ corresponds to no number z in the plane, we correspond it with ∞. Of course, the system consisting of $\mathbf{C}$ adjoined with the "ideal point" ∞ is called the **extended complex-number system.**

This way of corresponding or mapping the complex numbers onto a sphere—north pole $(0, 0, 1)$ excluded—is called a **stereographic projection.**



(a) Unit circle **(b)** Unit sphere

Solve the quadratic equation $z^2 + (1-i)z - 3i = 0$.

Solution $a = 1,\ b = 1 - i$, and $c = -3i$ we have

$$z = \frac{-(1-i) + [(1-i)^2 - 4(-3i)]^{1/2}}{2} = \frac{1}{2}\left[-1 + i + (10i)^{1/2}\right]. \qquad ($$

To compute $(10i)^{1/2}$ $r = \sqrt{10},\ \theta = \pi/2$, and $n = 2,\ k = 0,\ k = 1$. The two square roots of $10i$ are:

$$w_0 = \sqrt{10}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = \sqrt{10}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{5} + \sqrt{5}i$$

and

$$w_1 = \sqrt{10}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = \sqrt{10}\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right) = -\sqrt{5} - \sqrt{5}i.$$

Therefore,

$$z_1 = \frac{1}{2}\left[-1 + i + \left(\sqrt{5} + \sqrt{5}i\right)\right] \quad \text{and} \quad z_2 = \frac{1}{2}\left[-1 + i + \left(-\sqrt{5} - \sqrt{5}i\right)\right].$$

These solutions of the original equation, written in the form $z = a + ib$, are

$$z_1 = \frac{1}{2}\left(\sqrt{5} - 1\right) + \frac{1}{2}\left(\sqrt{5} + 1\right)i \quad \text{and} \quad z_2 = -\frac{1}{2}\left(\sqrt{5} + 1\right) - \frac{1}{2}\left(\sqrt{5} - 1\right)i.$$

$$e^z = e^{x+iy} = e^x(\cos y + i\sin y).$$

$$e^{z_1}e^{z_2} = e^{z_1+z_2}.$$

$$z = re^{i\theta}.$$

This convenient form is called the **exponential form** of a complex number z. For example, $i = e^{\pi i/2}$ and $1 + i = \sqrt{2}e^{\pi i/4}$. Also, the formula for the n nth roots of a complex number,

$$z^{1/n} = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}, k = 0, 1, 2, \ldots, n-1.$$